

Cloud-Enabled Fractional Hybrid Physics-Informed Neural Operator for Financial Derivative Pricing

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Abstract

We propose CF-HPINO, a cloud-enabled hybrid Fourier neural operator with physics-informed losses for Black-Scholes, fractional, and Merton option surfaces, trained on synthetic data and real SPY quotes. Compared to PINN [1] and PINN-FNO hybrids [4], CF-HPINO adds fractional/Merton physics, market supervision, and open code. Synthetic rel-L2 ~ 0.68 vs baselines ~ 0.99 ; SPY val rel-L2 4.29, quote RMSE \$379.

1. Introduction

Recent work applies physics-informed neural networks (PINNs) to option pricing: Dhiman and Hu [1] (arXiv:2312.06711); Bansal et al. [2] on PIDEs (Applied Soft Computing); Gatta et al. [3] on American options (ScienceDirect); Elbayed and Qadi el Idrissi [4] on hybrid PINNs-FNO; Chen et al. [5] on barrier options; Lee et al. [6] DeepONet bond pricing; Feng et al. [7] hybrid neural operators for SPDE pricing; Hainaut and Casas [8] Heston PINNs; illiquid jump markets [9]; fractional BS [10,14]; regime switching PIRL [11]; American jump-diffusion under data scarcity [12]; implied vol operators [13]; KAN finance networks [15]. CF-HPINO extends [4] with fractional/Merton losses, attention fusion, and Yahoo SPY data.

2. Methodology

Architecture: FNO backbone + physics MLP + attention fusion (cf. [4,7,14]). Loss: data + PDE residual + boundary + market quote terms following [1,3,raissi]. Models: Black-Scholes, fractional Caputo BS [10], Merton PIDE [2,9,12]. Data: synthetic surfaces; SPY full chain (5,283 quotes, 33 expiries). Training: Adam, EMA, curriculum; splits by parameter vector or expiry [11].

3. Results

Synthetic (held-out theta): CF-HPINO rel-L2 0.68 (3 epochs) vs PINN/FNO 0.99. Real SPY (40 epochs, CPU): val rel-L2 4.29; test rel-L2 4.64; quote RMSE \$379 ($\sim 51\%$ spot). Learnability confirmed but below desk accuracy in [8,11,13].

4. Discussion

CF-HPINO is closest to hybrid PINN-FNO [4] and SPDE operators [7]. ScienceDirect anchors include [3,5,11,12,14]. Limits: European focus; CPU training; grid-based market quotes. Future: Heston [8], regime switching [11], GPU full physics.

5. Conclusion

CF-HPINO delivers an open hybrid physics-informed neural operator aligned with 2023-2026 PINN/operator finance literature [1-15], with preliminary SPY validation.

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